

|                                  |                                 |
|----------------------------------|---------------------------------|
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| Batch :- B1                      | Class :- SYIT                   |
| Date of Assignment :- 18-08-2026 | Date of Submission :-18-08-2026 |

# Practical 19

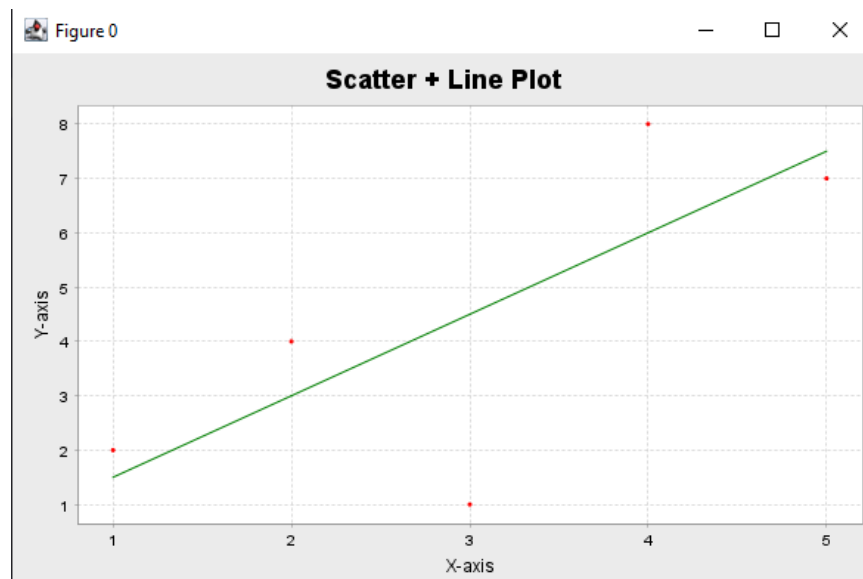
Code -

```

1 import breeze.linalg._
2 import breeze.plot._
3
4 object Jasad extends App {
5   val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Sample X-axis values
6   val yscat = DenseVector(2.0, 4.0, 1.0, 8.0, 7.0) // Y values for scatter
7   val yline = DenseVector(1.5, 3.0, 4.5, 6.0, 7.5) // Y values for line plot
8   val fig = Figure() // Create a new figure
9   val plt = fig.subplot(0)
10  plt += plot(x, yscat, '.', colorcode = "Red")
11  plt += plot(x, yline, '-', colorcode = "Green")
12  plt.title = "Scatter + Line Plot"
13  plt.xlabel = "X-axis"
14  plt.ylabel = "Y-axis"
15  fig.refresh()
16
17  println("Jasad S024")
18 }

```

Output -



# Practical 20

Code -

```
build.sbt  d20.scala x Iris.csv
1 import scala.io.Source
2 object d20 extends App {
3     val file = Source.fromResource("iris.csv")
4     val data = file.getLines()
5         .drop(1)
6         .map(line => line.split(",")(0).toDouble)
7         .toList
8
9     file.close()
10
11     val count = data.length
12     val mean = data.sum / count
13     val min = data.min
14     val max = data.max
15
16     val variance = data.map(x => math.pow(x - mean, 2)).sum / count
17     val std = math.sqrt(variance)
18
19     println(s"Count: $count")
20     println(f"Mean: $mean%.2f")
21     println(f"Minimum: $min%.2f")
22     println(f"Maximum: $max%.2f")
23     println(f"Standard Deviation: $std%.2f")
24     println("Jasad s024")
25 }
```

Output -

```
Count: 150
Mean: 75.50
Minimum: 1.00
Maximum: 150.00
Standard Deviation: 43.30
Jasad s024

Process finished with exit code 0
```

# Practical 21

Code -

```
ThisBuild / scalaVersion := "2.13.12"

libraryDependencies += Seq(
  "org.apache.spark" %% "spark-core" % "3.5.1",
  "org.apache.spark" %% "spark-sql" % "3.5.1"
)

lazy val root = (project in file("."))
  .settings(
    name := "Jasad1"
  )
```

```
import org.apache.spark.sql.SparkSession

object Jasad extends App {
  val s = SparkSession.builder()
    .appName("JoinTwoCSV")
    .master("local[*]")
    .getOrCreate()
  val d1 = s.read
    .option("header", "true")
    .csv("file1.csv")
  val d2 = s.read
    .option("header", "true")
    .csv("file2.csv")
  val j = d1.join(d2, Seq("id"), "inner")
  j.show()
  j.write.option("header", "true")
  s.stop()
}
```

| build.sbt |         | Jasad.scala | file1.csv × | file2.csv |
|-----------|---------|-------------|-------------|-----------|
| 1         | id,name |             |             |           |
| 2         | 1,rohit |             |             |           |
| 3         | 2,priya |             |             |           |

| build.sbt |          | Jasad.scala | file1.csv | file2.csv × |
|-----------|----------|-------------|-----------|-------------|
| 1         | id,score |             |           |             |
| 2         | 2,90     |             |           |             |
| 3         | 1,70     |             |           |             |

```

Run  Jasad x
26/08/18 11:55:19 INFO DAGScheduler: Job 3 finished. Show at Jasad.scala:13, took 0.214514 s
26/08/18 11:55:20 INFO CodeGenerator: Code generated in 31.7345 ms
+---+-----+-----+
| id| name|score|
+---+-----+-----+
| 1|rohit| 70|
| 2|priya| 90|
+---+-----+-----+

26/08/18 11:55:20 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/08/18 11:55:20 INFO SparkUI: Stopped Spark web UI at http://DESKTOP-2JJMAUE:4040
26/08/18 11:55:20 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/08/18 11:55:20 INFO MemoryStore: MemoryStore cleared
26/08/18 11:55:20 INFO BlockManager: BlockManager stopped
26/08/18 11:55:20 INFO BlockManagerMaster: BlockManagerMaster stopped
26/08/18 11:55:20 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator

```